

Grade 11 Mathematics

Equations & Inequalities & Exponents & Surds

Task: Task 7

Term: Term 3 2026

Time: 60 minutes

Total Marks: 11

Examiner: AE Killer

Moderator: M Visser

QUESTION 1

1.1 Solve for x : $2x - 5 = 0$ (2)

1.2 Sipho borrows R500 000 at 6% p.a. compound interest.

1.2.1 Calculate the amount owed after 5 years. (2)

1.2.2 How many years to pay back at current rate? (3)

1.3 Factorize: $x^2 - 3x - 10$ (2)

[9]

QUESTION 2

2.1 Simplify: $2^3 \cdot 2^2$ (2)

[2]

TOTAL [11]

MARKING GUIDELINE

QUESTION 1		
1.1	$2x - 5 = 0$	✓ standard form
	$x = 2,5$	✓ answer (2)
1.2.1	$A = P(1+i)^n$	✓ formula
	$A = 500000(1,06)^5 = 669\,113,35$	✓ substitution and calculation (2)
1.2.2	$1000000 = 500000(1,06)^n$	✓ set up equation
	$2 = (1,06)^n$	✓ simplify
	$n = \frac{\ln 2}{\ln 1,06} \approx 12,5$	✓ logarithmic solution (3)
1.3	$(x-5)(x+2)$	✓ ✓ factors (2)

QUESTION 2		
2.1	$2^5 = 32$	✓ ✓ answer (2)

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TOTAL [11]

COGNITIVE LEVEL ANALYSIS GRID

QUESTION	Knowledge	Routine	Complex	Problem-Solving	Total
QUESTION 1	0	6	3	0	9
QUESTION 2	2	0	0	0	2
TOTAL	2	6	3	0	11
FET TARGET %	20%	35%	30%	15%	—
THIS TASK %	18%	55%	27%	0%	100%